

Synthesis of 4-Substituted Phosphacoumarins via Cross-Coupling of 4-Tosylphosphacoumarins with Organozinc Reagents

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Abstract: A series of 4-substituted phosphacoumarin derivatives with potential biological and pharmacological activity has been synthesized through Negishi-type cross-coupling reactions of 4-tosylphosphacoumarins with organozinc reagents.

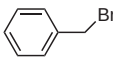
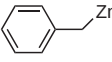
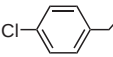
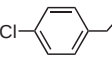
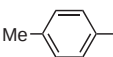
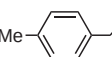
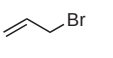
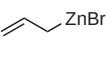
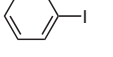
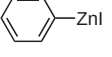
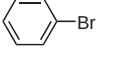
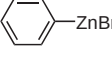
Key words: phosphacoumarin, Negishi-type cross-coupling, organozinc reagents

Coumarin and its derivatives form an important class of benzopyrones found in nature. They are a structural subunit in many complex natural products and have shown various biological activities, such as antitumor,¹ anti-HIV,² antioxidation,³ tumor necrosis factor- α (TNF- α) inhibition,⁴ antimicrobial activity,⁵ serine protease inhibition,⁶ and anticancer activity.⁷ The widespread biological activities of coumarin derivatives have aroused great interest in the area of synthetic chemistry and pharmacology. At the same time, organophosphorus compounds are of great importance in the development of chemistry and pharmacology.⁸ Considering the similarity of phosphonic acid derivatives to the naturally occurring carboxylic acid derivatives in biology and pharmacology, we have recently designed and synthesized analogues of coumarins and phosphacoumarins, by introducing the phosphorus atom into the ring of coumarin derivatives, and the preliminary bioassay showed that the designed phosphacoumarins demonstrated moderate to good inhibitory activity against SHP-1, a kind of protein tyrosine phosphatase.⁹ Encouraged by these results, we decided that it was necessary to enlarge the library of phosphacoumarins so that molecules with more potent biological activities can be found.

Organozinc reagents are widely used in carbon–carbon bond formation through the Negishi cross-coupling reaction¹⁰ because they are easily accessible and tolerant of many functional groups such as esters and lactones compared to other organometallic reagents, such as Grignard reagents and organolithiums.¹¹ We initiated a program to synthesize a series of 4-alkyl and aryl phosphacoumarins through the palladium-catalyzed cross-coupling reactions of 4-tosylphosphacoumarins with organozinc reagents.

Organozinc reagents **2** were prepared by the reaction of alkyl or aryl halides with an excess of zinc powder in the presence of a catalytic amount of TMSCl by a reported procedure (Table 1).¹²

Table 1 Preparation of Organozinc Reagents

$\text{Zn (powder)} + \text{RX} \xrightarrow[55 \text{ or } 65^\circ\text{C}]{\text{TMSCl (3 mol\%), THF}} \text{RZnX}$				
Entry	RX 1	RZnX 2	Temp (°C)	Time (h)
1			55	10
2			55	10
3			55	10
4			55	10
5			65	24
6			65	24



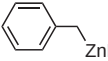
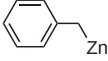
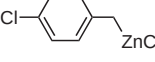
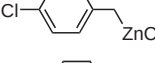
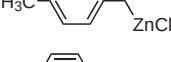
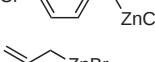
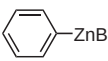
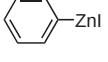
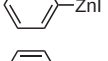
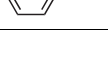
Scheme 1 Negishi cross-coupling reactions of 4-substituted phosphacoumarins with organozinc reagents. *Reagents and conditions:* R^2ZnX (1.5 equiv), $\text{Pd}(\text{PPh}_3)_4$ (5 mol%), PPh_3 (5 mol%), anhyd THF, N_2 , 55 °C, 2–8 h.

4-Tosylphosphacoumarin derivatives were synthesized from the reaction of phosphachroman-2,4-diones and 4-OH phosphacoumarins with *p*-toluenesulfonyl chloride.⁹ Reaction of 4-tosylphosphacoumarins with organozinc reagents catalyzed by 5% mol $\text{Pd}(\text{PPh}_3)_4$ and 5% mol PPh_3 under a nitrogen atmosphere (Scheme 1) yielded the corresponding Negishi cross-coupling products in 64–85% isolated yields (Table 2). The Negishi cross-coupling reactivity was affected by the electronic nature of the substrate. For example, compounds **3** containing an electron-withdrawing group (R^1) gave higher yields than those

containing an electron-donating group (Table 2, compare entries 4 with 6 and 10 with 11). Correspondingly, organozinc reagents with an electron-withdrawing group provided higher yields (Table 2, compare entries 2 with 3 and 4 with 5). In addition, the organozinc reagent derived from phenyl iodide gave higher yields than that from phenyl bromide (Table 2, entries 8–11).

In summary, various 4-substituted phosphacoumarin derivatives have been synthesized through the cross-coupling reactions of 4-tosylphosphacoumarins with organozinc reagents (Negishi-type). The method provides a more convenient and efficient strategy for the introduction of aryl and alkyl substituents onto phosphacoumarins relative to the Suzuki-type cross-coupling reaction.⁹ An enriched library of coumarins, phosphacoumarins, and organophosphorus compounds, enhances the possibility of finding small molecules with biological and pharmacological activities.

Table 2 Negishi Cross-Coupling Reaction of 4-Tosylphosphacoumarins with Organozinc Reagents¹³

Entry	Compd	R ¹	R ² ZnX	Time (h)	Yield (%) ^a
1	2a	H		5	72
2	2b	Cl		5	75
3	2c	Cl		5	82
4	2d	CN		8	83
5	2e	CN		8	75
6	2f	Me		8	79
7	2g	Cl		5	85
8	2h	Cl		2	64
9	2h	Cl		2	78
10	2i	CN		2	84
11	2j	Me		2	75

^a Isolated yield.

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- (13) Analytical data of 4-benzyl-2-ethoxy-3-phenylphosphacoumarin (**4a**) and 4-allyl-3-(4'-chlorophenyl)-2-ethoxyphosphacoumarin (**4g**) (Figure 1).

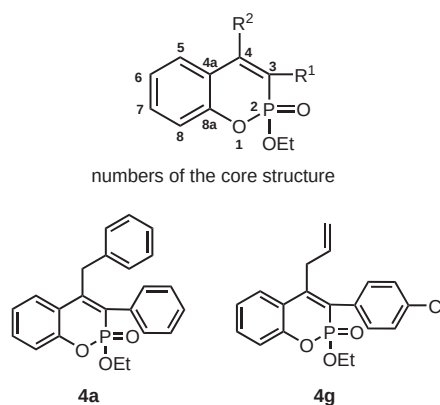


Figure 1

4a White foam; yield: 72%. ¹H NMR (300 MHz, CDCl₃): δ = 1.17 (t, ³J = 7.2 Hz, 3 H, CH₂CH₃), 4.07 (m, 4 H, CH₂CH₃, PhCH₂), 7.12–7.40 (m, 14 H, ArH). ¹³C NMR (75 MHz, CDCl₃): δ = 16.4 (d, ³J_{P-C} = 5.7 Hz, POCH₂CH₃), 36.0 (d, ³J_{P-C} = 15.1 Hz, PhCH₂), 63.9 (d, ²J_{P-C} = 6.5 Hz, POCH₂CH₃), 119.4–138.4 (ArC), 146.6 (C-4), 150.9 (d, ²J_{P-C} = 7.2 Hz, C-8a). ³¹P NMR (125 MHz, CDCl₃): δ = 8.57. HRMS: *m/z* calcd for C₂₃H₂₂PO₃ [M + H]⁺, 377.1307; found: 377.1304.

4g Yellowish viscous oil; yield: 85%. ^1H NMR (300 MHz, CDCl_3): δ = 1.17 (t, 3J = 7.2 Hz, 3 H, CH_2CH_3), 3.32 (m, 2 H, $\text{CH}_2=\text{CHCH}_2$), 4.08 (m, 2 H, OCH_2CH_3), 4.95–5.12 (dd, 2 H, $\text{CH}_2=\text{CHCH}_2$), 5.84 (m, 1 H, $\text{CH}_2=\text{CHCH}_2$), 7.18–7.57 (m, 8 H, ArH). ^{13}C NMR (75 MHz, CDCl_3): δ = 16.4 (d, $^3J_{\text{P-C}}$ = 6.5 Hz, POCH_2CH_3), 34.2 (d, $^3J_{\text{P-C}}$ = 15.10 Hz, $\text{CH}_2=\text{CHCH}_2$), 63.8 (d, $^2J_{\text{P-C}}$ = 6.5 Hz, POCH_2CH_3), 117.8–134.5 (ArC, $\text{CH}_2=\text{CH}$), 147.2 (C-4), 150.8 (C-8a). ^{31}P NMR (125 MHz, CDCl_3): δ = 8.13. HRMS: m/z calcd for $\text{C}_{19}\text{H}_{19}\text{PO}_3\text{Cl}$ [$\text{M} + \text{H}$] $^+$: 361.0760; found: 361.0753.

Organozinc Reagents; General Procedure

Zn powder was treated with 1 M HCl for 10 min, washed with acetone and then dried under vacuum before use. A mixture of Zn powder (3.0 mmol) and TMSCl (0.5 mL) in THF (10 mL) was stirred under a nitrogen atmosphere at r.t. for 20 min, and then heated to 55–65 °C (Table 1). Alkyl or aryl halide (**3**, 2.0 mmol) in anhyd THF (2 mL) was added dropwise, and the reaction was maintained at this

temperature for 10–24 h. The resulting solution was cooled to r.t., and the clear liquid could be directly used for the following reaction.

Negishi Cross-Coupling Reaction of Organozinc Reagents with 4-Tosylphosphacoumarins General Procedure

A mixture of 4-tosylphosphacoumarin (**1**) (1.0 mmol), $\text{Pd}(\text{PPh}_3)_4$ (0.05 mmol), PPh_3 (0.05 mmol) and the organozinc reagent prepared above in anhyd THF (10 mL) was heated to 55 °C under a nitrogen atmosphere. After 2–8 h, the reaction mixture was cooled to r.t. After the solvent was removed under reduced pressure, H_2O (10 mL) was added to the residue, the solution was extracted with CH_2Cl_2 (3×5 mL), and the combined organic phase was dried over anhyd Na_2SO_4 . After removal of the solvent, the residue was purified by column chromatography on silica gel (PE–EtOAc, 2:1) to give product **2**.